

Spiking Neural Network (SNN) Robustness Benchmark

Box Surrogate (β) sweeps under Data Heterogeneity and Attacks

ByzFL SNN Benchmark

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1 Introduction and Experimental Settings

This report documents the robustness sweeps of Spiking Neural Networks (SNN) using the Box surrogate gradient. The width parameter β is swept across values: $\beta \in \{0.25, 0.5, 0.75, 1.0, 1.25, 2.0\}$.

The performance of the model is evaluated in federated learning under varying data heterogeneity (non-IID) levels, under Byzantine attacks, using different robust aggregators. For each configuration, the Spiking Neural Network results are compared side-by-side with a standard (non-spiking) Convolutional Neural Network baseline (`cnn_mnist`) evaluated under the same conditions.

1.1 Parameter Configurations

Table 1 details the parameters, hyperparameters, and algorithms used in this sweep.

Table 1: Experiment Parameters and Hyperparameters

Component / Parameter	Configuration Value
Dataset	MNIST (60,000 train, 10,000 test)
Model Structure	Spiking Convolutional Neural Network (<code>cnn_mnist_snn</code>)
Loss Function	Rate-coded Cross-Entropy Loss (<code>ce_rate_loss</code>)
Optimizer	SGD
Learning Rate	0.10
Momentum	0.9
Weight Decay	0.0001
Batch Size	128
SNN Encoding	Constant encoding, 10 time steps
SNN Decay (β_{leaky})	0.95
SNN Threshold	1.0 (fixed)
Training Steps	500
Training Algorithm	DSGD
Honest Nodes (N)	10
Byzantine Nodes (f)	$f \in \{0, 1, 2, 3, 4, 5\}$
Data Heterogeneity (γ)	$\gamma \in \{1.0, 0.66, 0.33, 0.0\}$ (from IID to extreme Non-IID)
Pre-aggregators	NNM, ARC
Aggregators Swept	Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean
Attacks Evaluated	Optimal ALIE, Sign Flipping, Optimal Inner Product Manipulation
Training Seeds	5 seeds (42 to 46)
Data Dist. Seed	42

2 Best Overall Performance (Worst-Case Across Attacks)

The heatmaps in this section display the worst-case test accuracy at convergence across all attacks achieved by selecting the best robust aggregator for each cell.

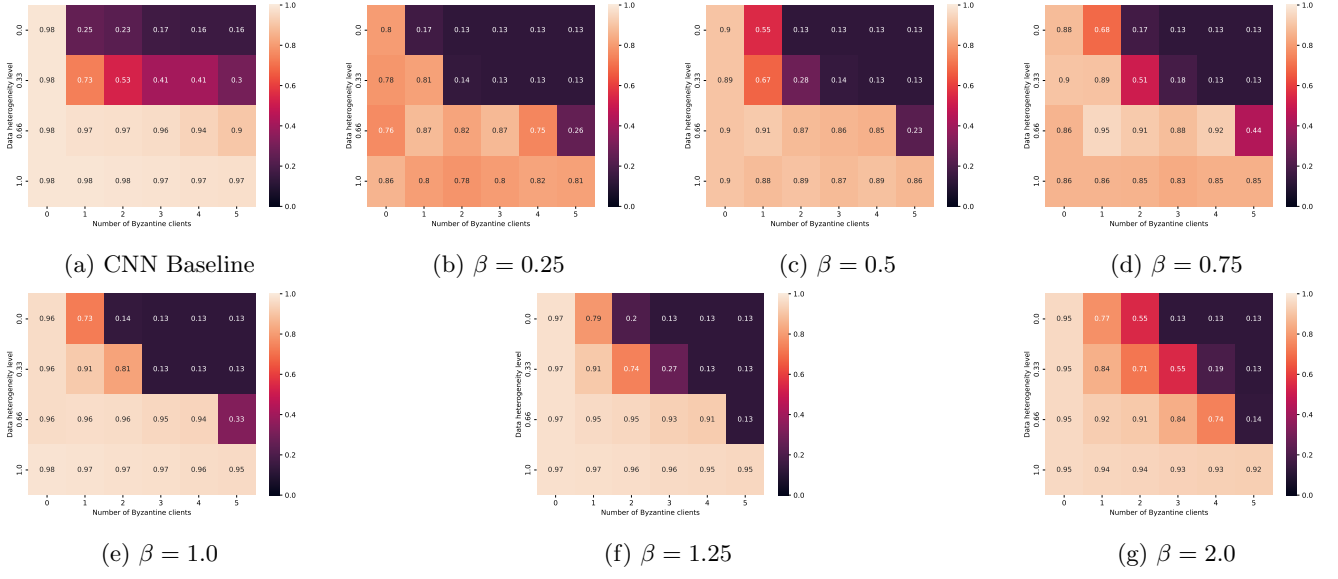


Figure 1: Best overall test accuracy heatmaps (worst-case across attacks) for different β parameter values.

3 Best Aggregator under Specific Attacks

This section presents heatmaps showing the best aggregator performance under each specific attack across all betas.

3.1 Optimal A Little Is Enough (ALIE) Attack

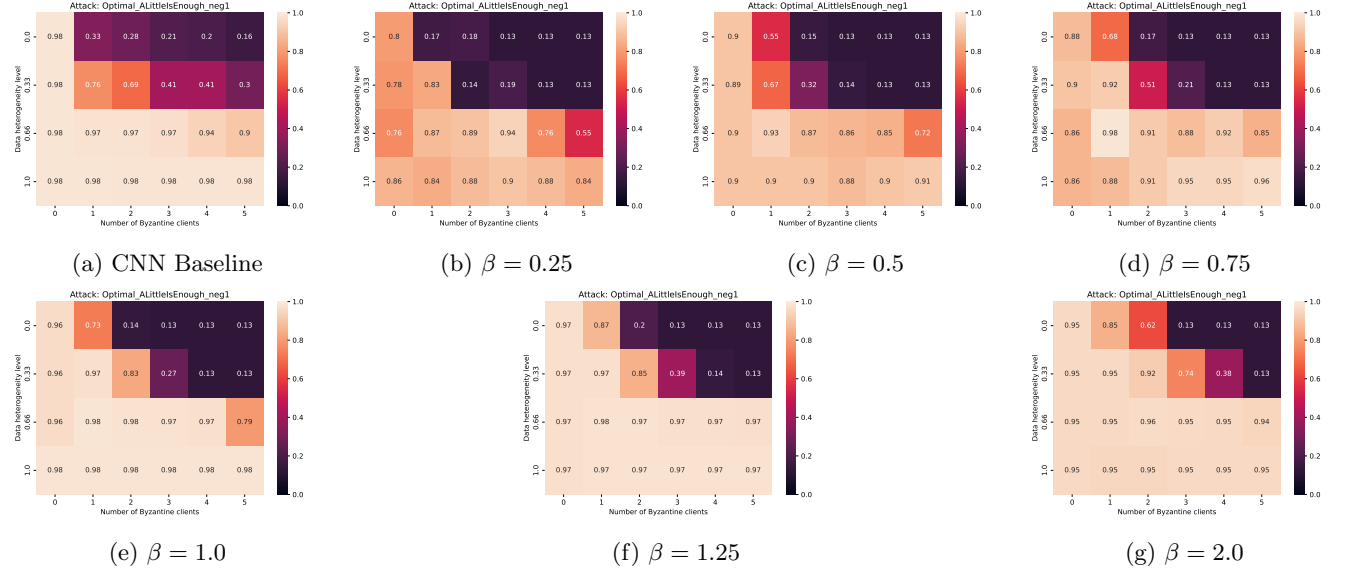


Figure 2: Best test accuracy heatmaps under the Optimal ALIE attack for different β parameter values.

3.2 Sign Flipping Attack

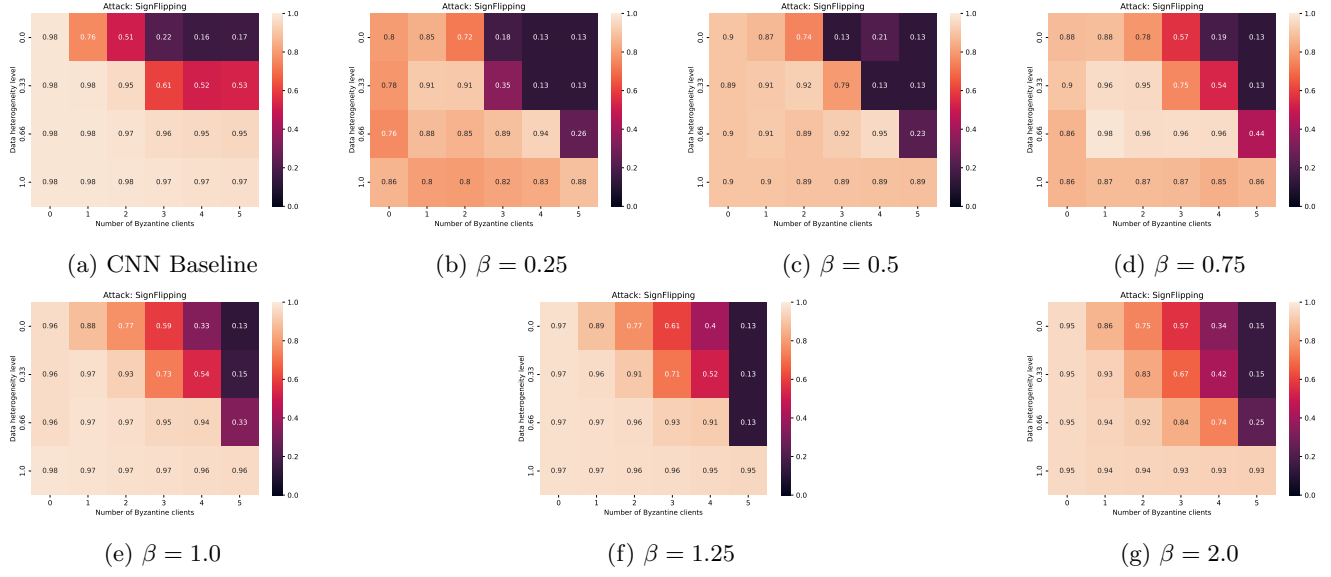


Figure 3: Best test accuracy heatmaps under the Sign Flipping attack for different β parameter values.

3.3 Optimal Inner Product Manipulation Attack

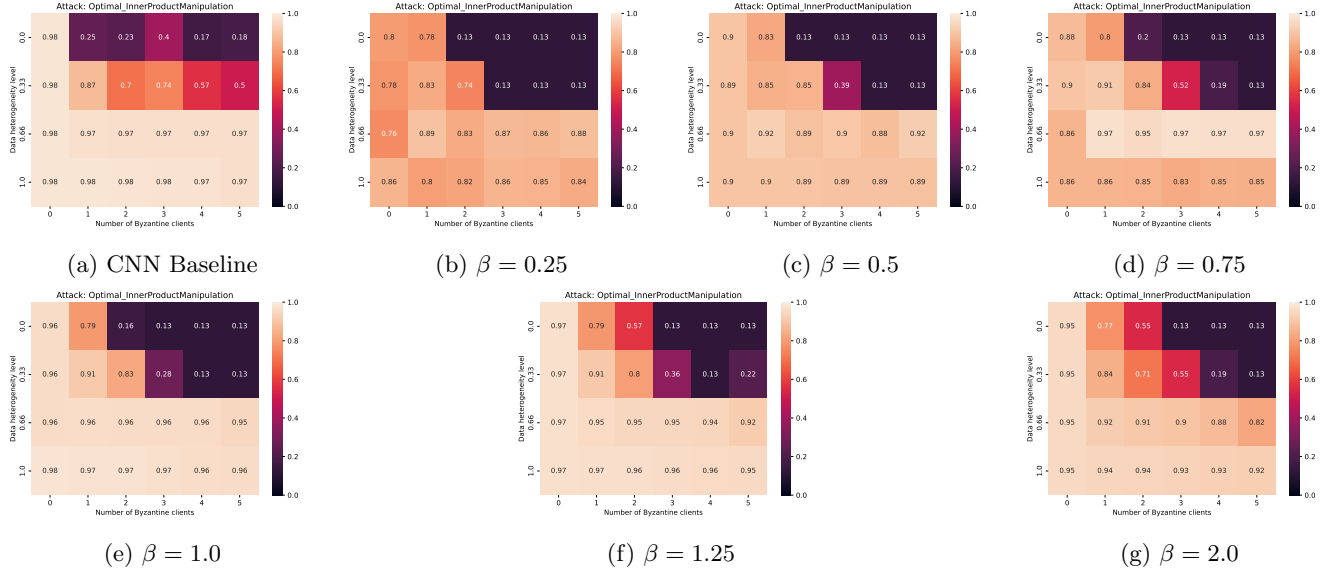


Figure 4: Best test accuracy heatmaps under the Optimal Inner Product Manipulation attack for different β parameter values.

4 Robustness of Fixed Aggregators across all Attacks

In this section, we analyze the performance of each specific aggregator (Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean) across all Byzantine attack settings for different width β values.

4.1 Centered Clipping (CC)

4.1.1 CC under Worst-Case across all Attacks

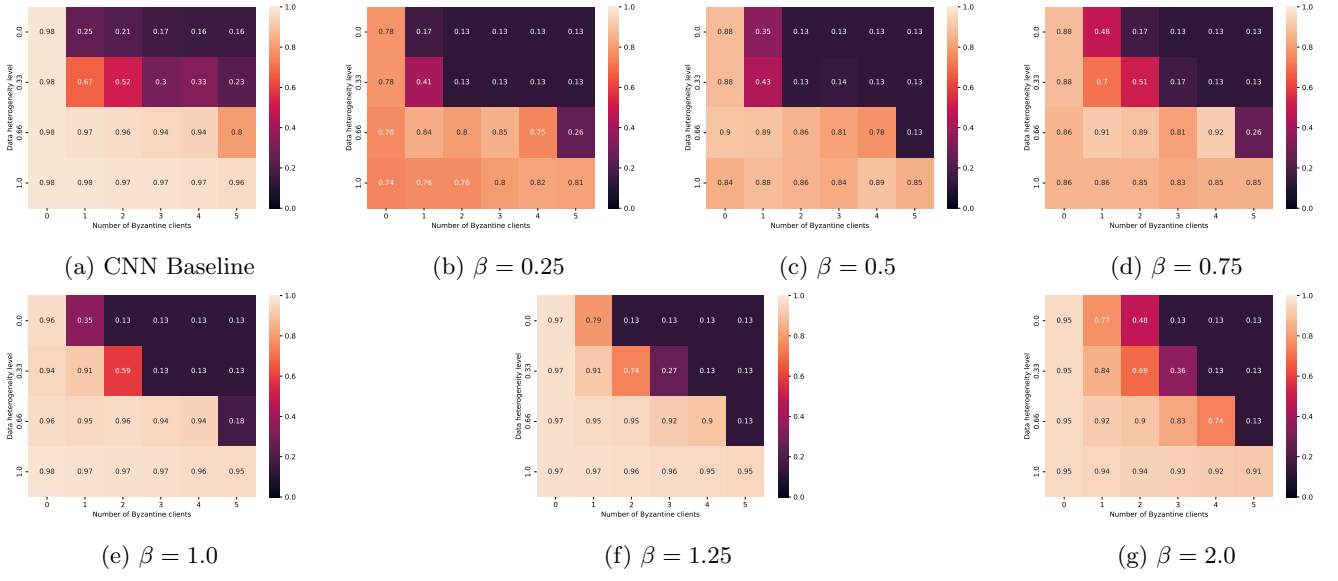


Figure 5: Centered Clipping performance under the worst-case across all attacks across β values.

4.1.2 CC under Optimal ALIE Attack

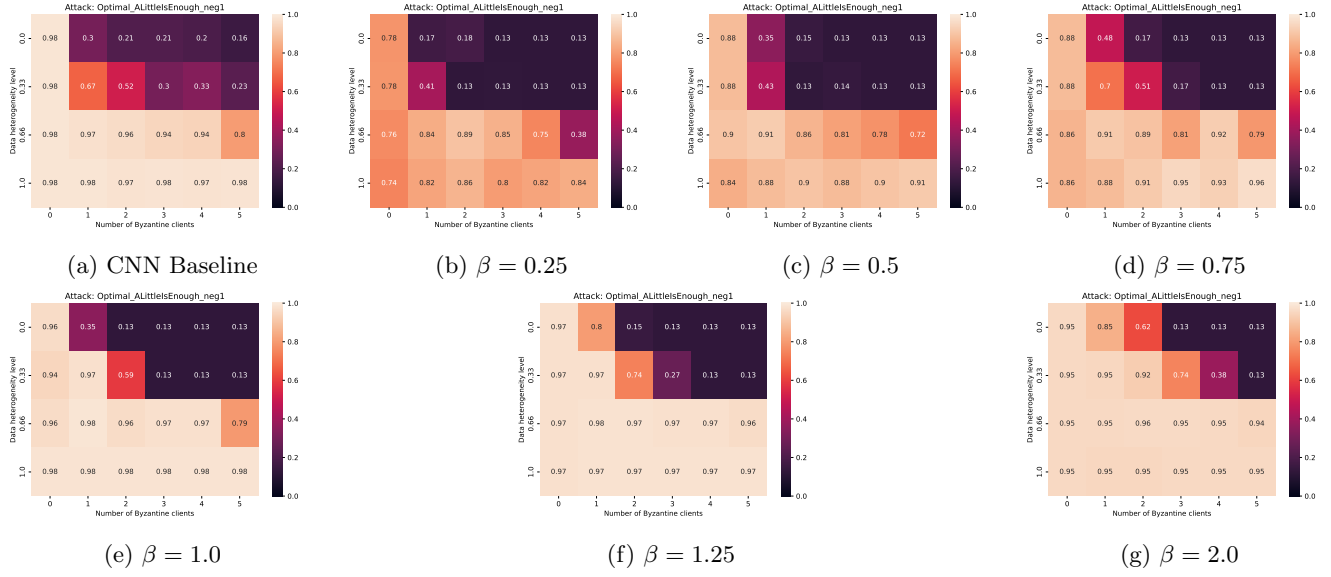


Figure 6: Centered Clipping performance under Optimal ALIE across β values.

4.1.3 CC under Sign Flipping Attack

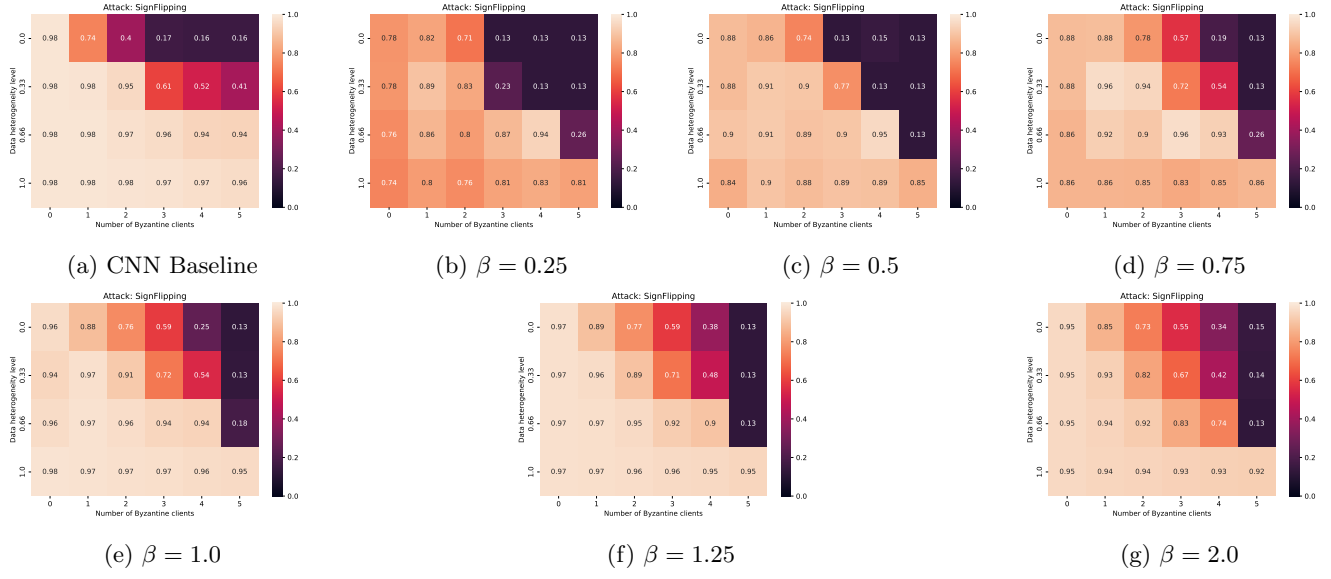


Figure 7: Centered Clipping performance under Sign Flipping across β values.

4.1.4 CC under Optimal Inner Product Manipulation Attack

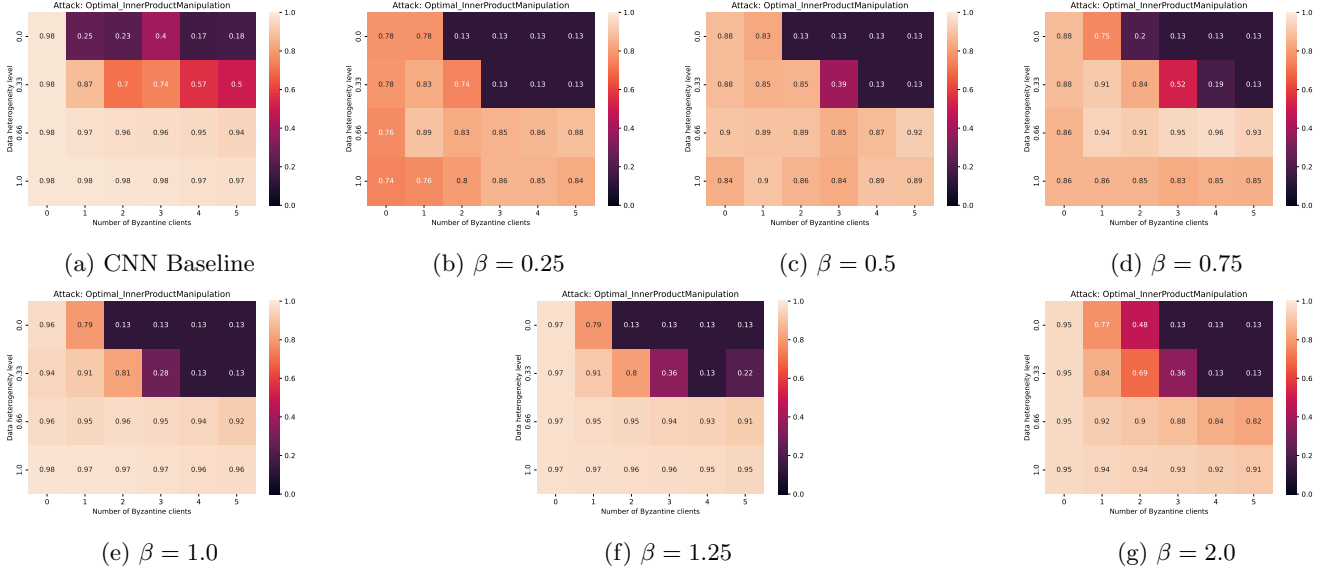


Figure 8: Centered Clipping performance under Optimal Inner Product Manipulation across β values.

4.2 Geometric Median (GM)

4.2.1 GM under Worst-Case across all Attacks

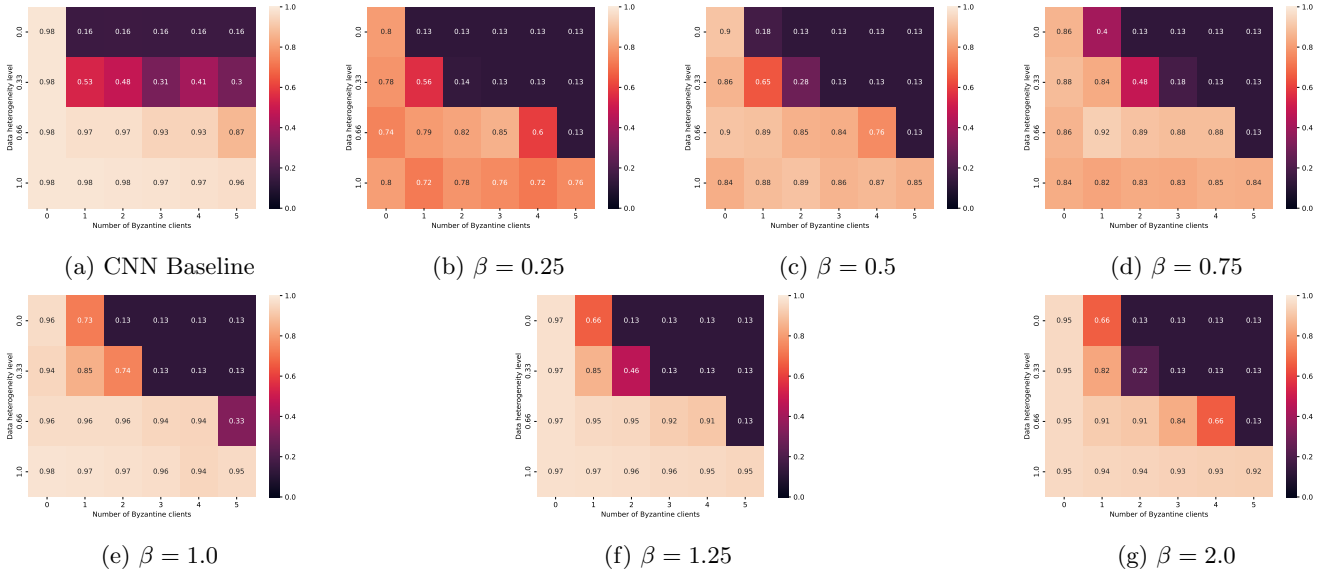


Figure 9: Geometric Median performance under the worst-case across all attacks across β values.

4.2.2 GM under Optimal ALIE Attack

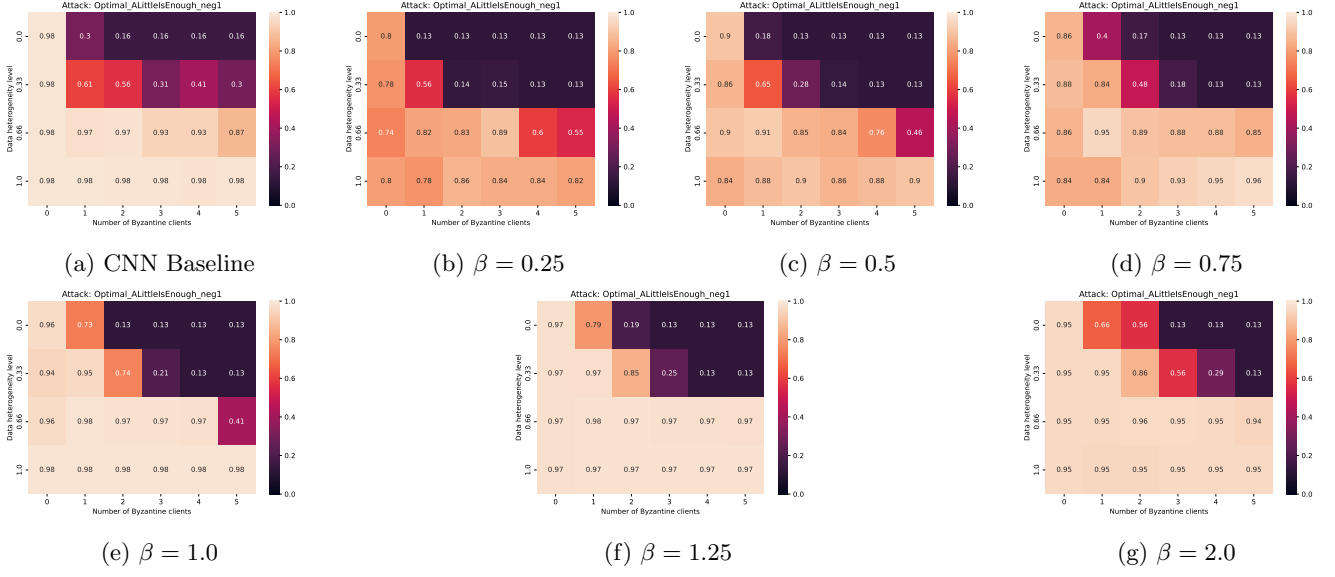


Figure 10: Geometric Median performance under Optimal ALIE across β values.

4.2.3 GM under Sign Flipping Attack

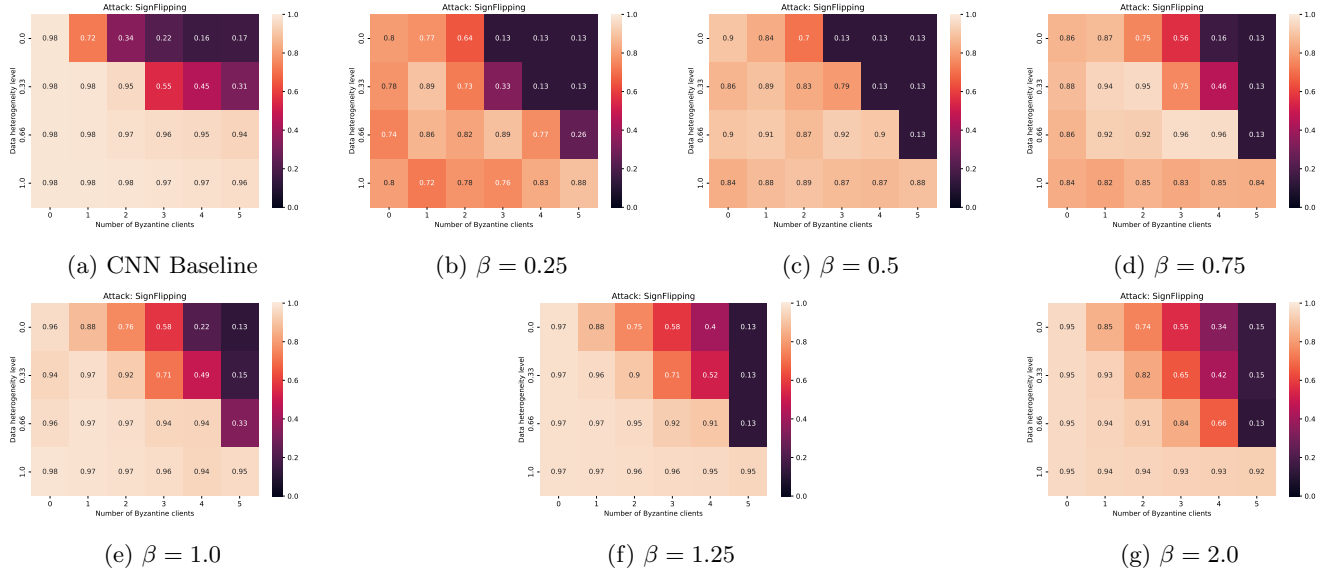


Figure 11: Geometric Median performance under Sign Flipping across β values.

4.2.4 GM under Optimal Inner Product Manipulation Attack

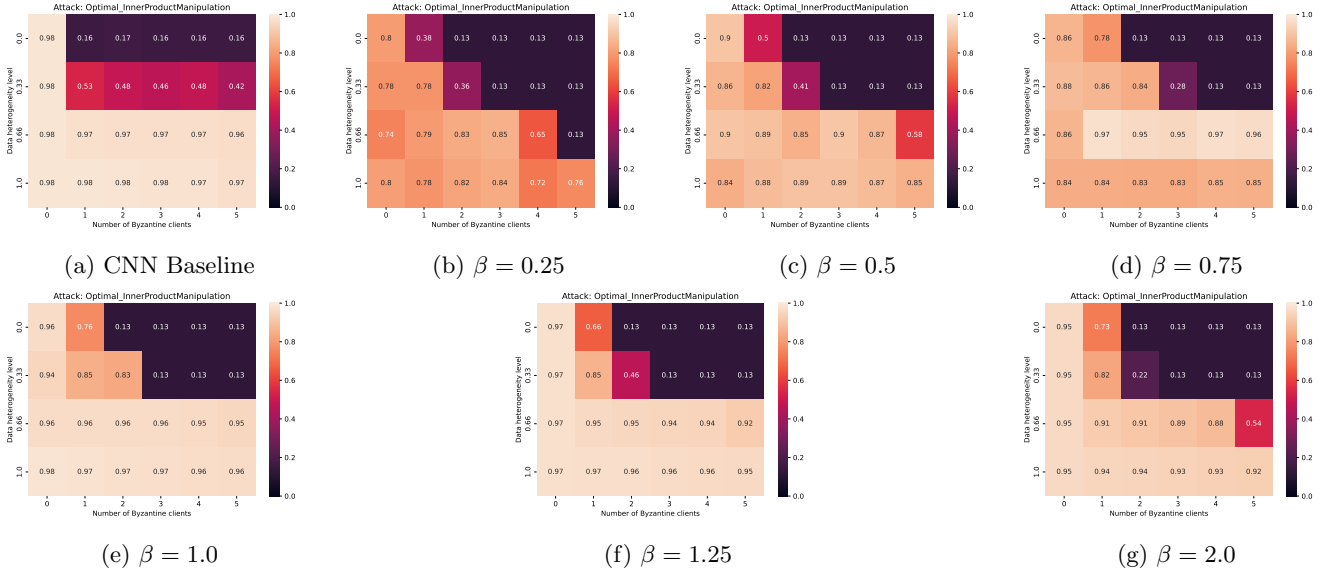


Figure 12: Geometric Median performance under Optimal Inner Product Manipulation across β values.

4.3 Multi-Krum (MK)

4.3.1 MK under Worst-Case across all Attacks

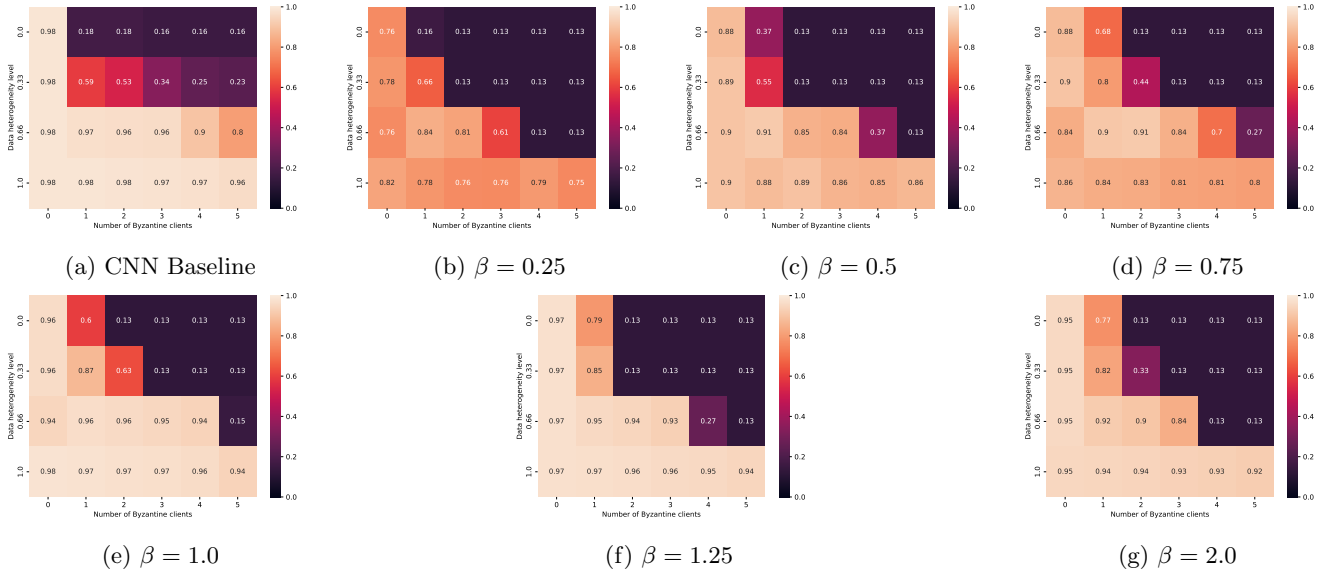


Figure 13: Multi-Krum performance under the worst-case across all attacks across β values.

4.3.2 MK under Optimal ALIE Attack

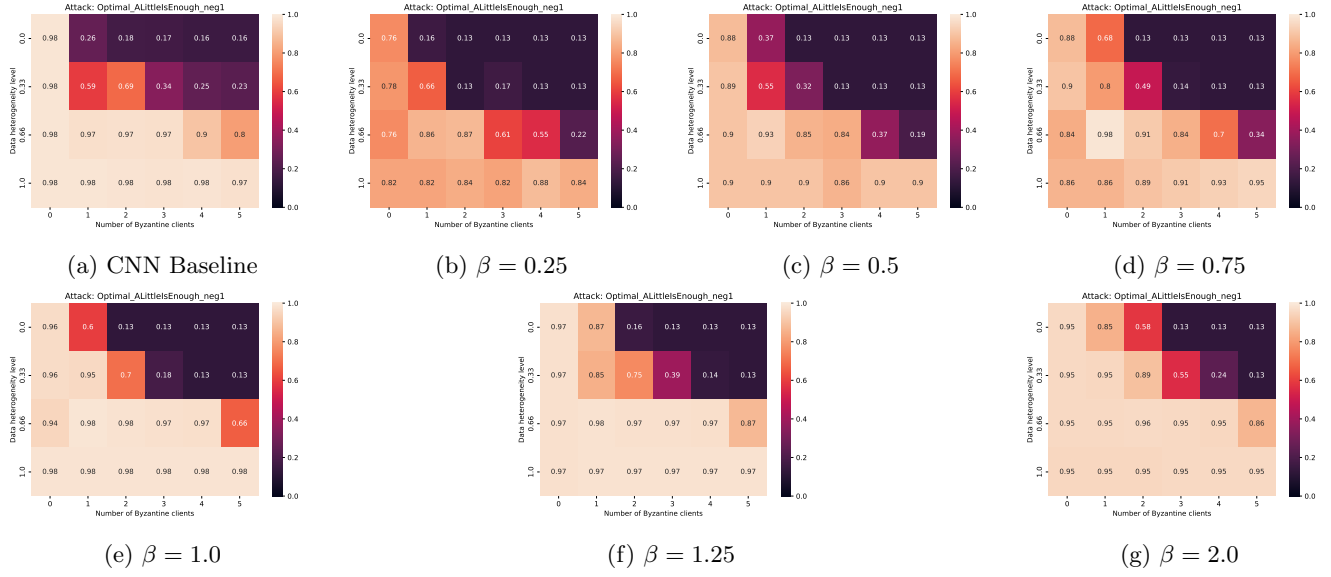


Figure 14: Multi-Krum performance under Optimal ALIE across β values.

4.3.3 MK under Sign Flipping Attack

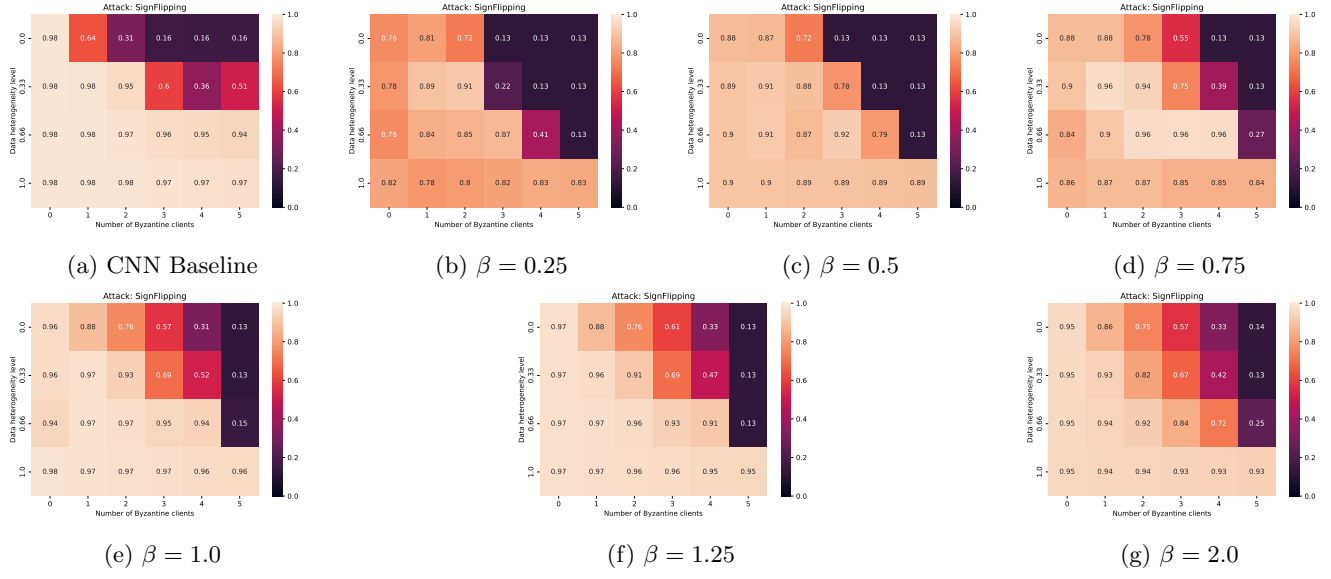


Figure 15: Multi-Krum performance under Sign Flipping across β values.

4.3.4 MK under Optimal Inner Product Manipulation Attack

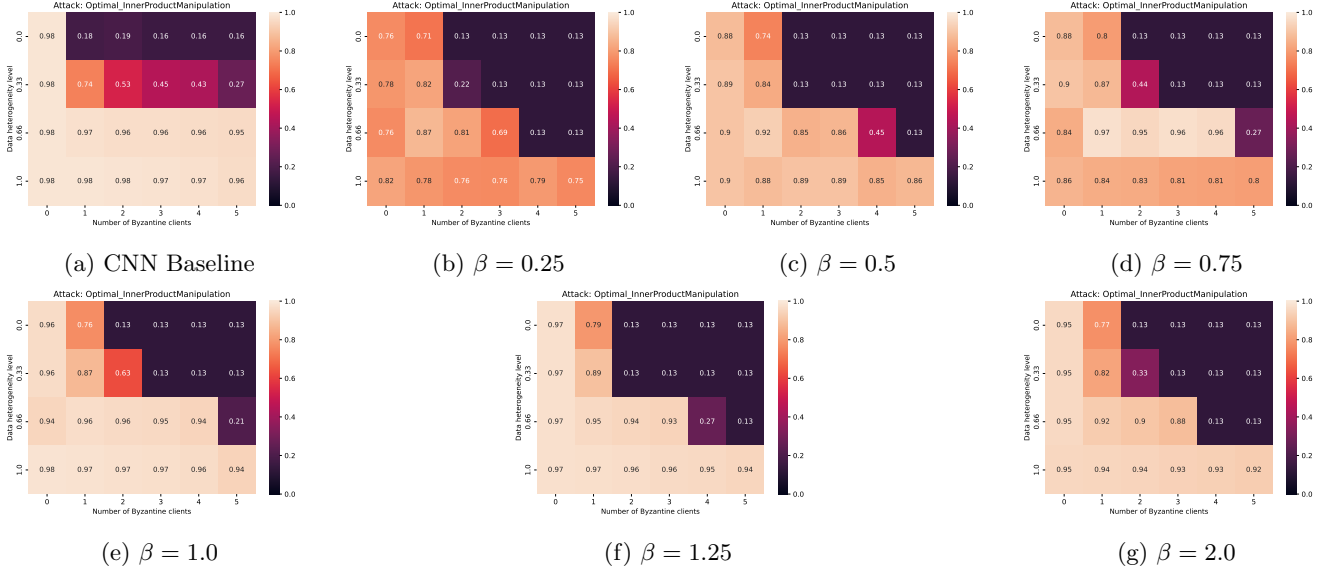


Figure 16: Multi-Krum performance under Optimal Inner Product Manipulation across β values.

4.4 Trimmed Mean (TM)

4.4.1 TM under Worst-Case across all Attacks

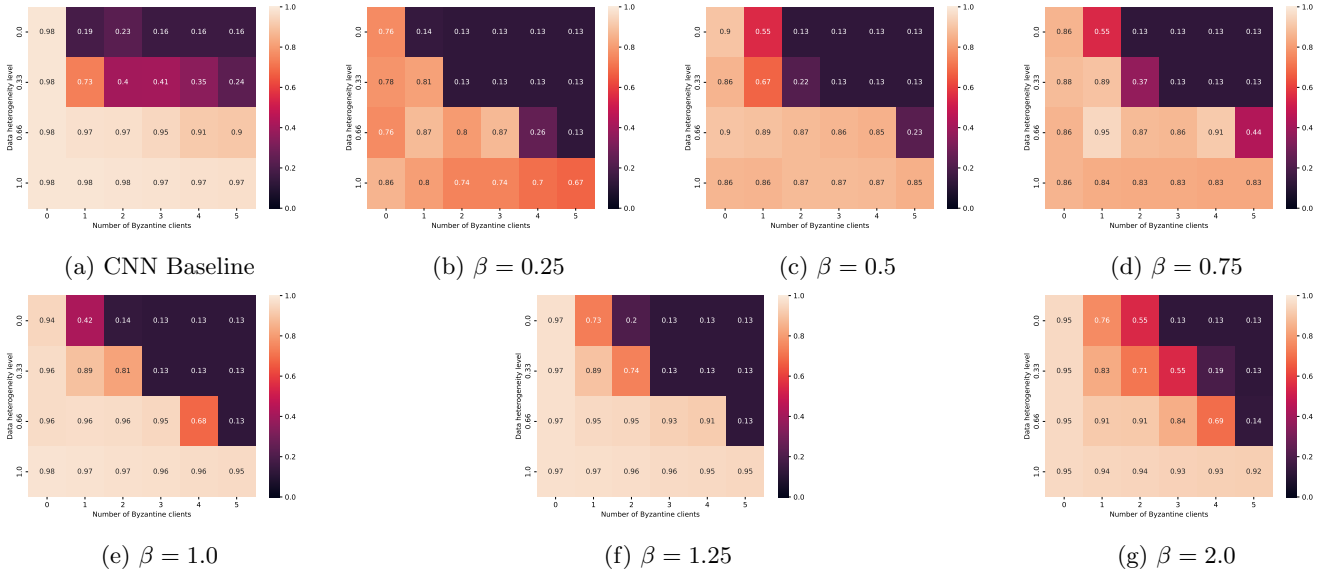


Figure 17: Trimmed Mean performance under the worst-case across all attacks across β values.

4.4.2 TM under Optimal ALIE Attack

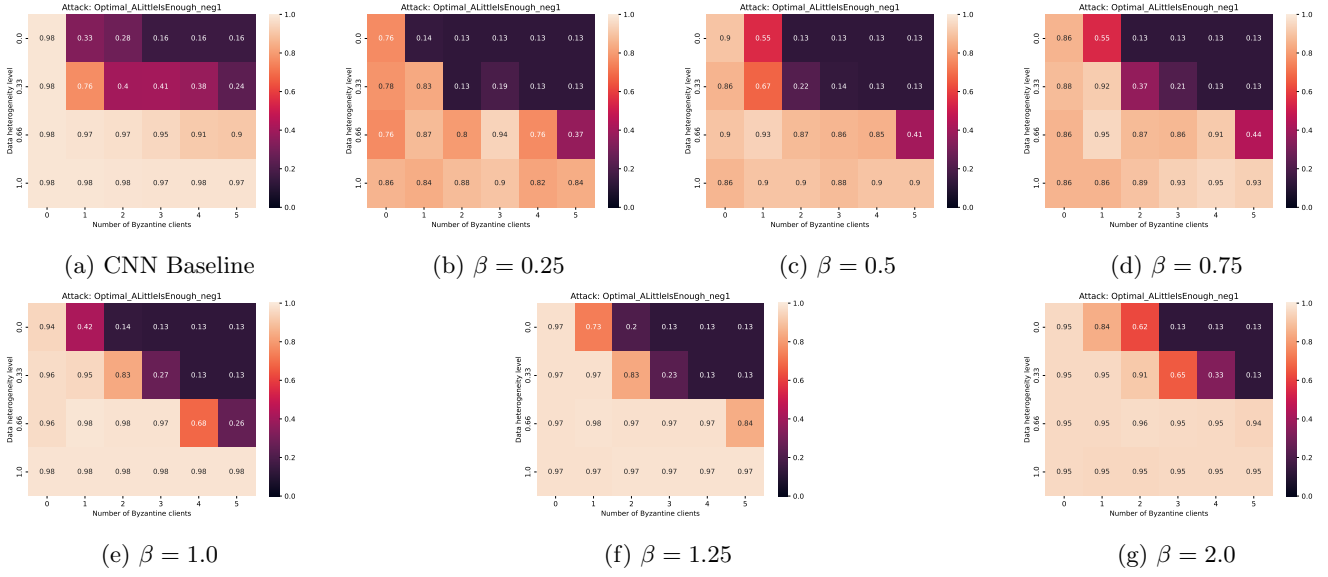


Figure 18: Truncated Mean performance under Optimal ALIE across β values.

4.4.3 TM under Sign Flipping Attack

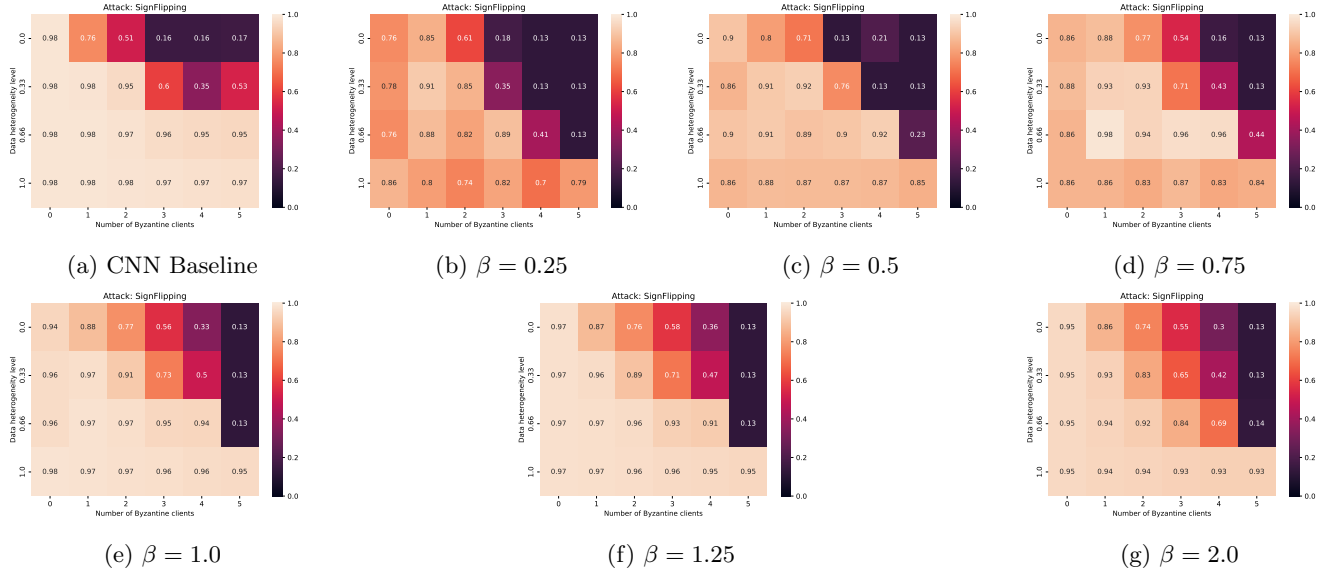


Figure 19: Truncated Mean performance under Sign Flipping across β values.

4.4.4 TM under Optimal Inner Product Manipulation Attack

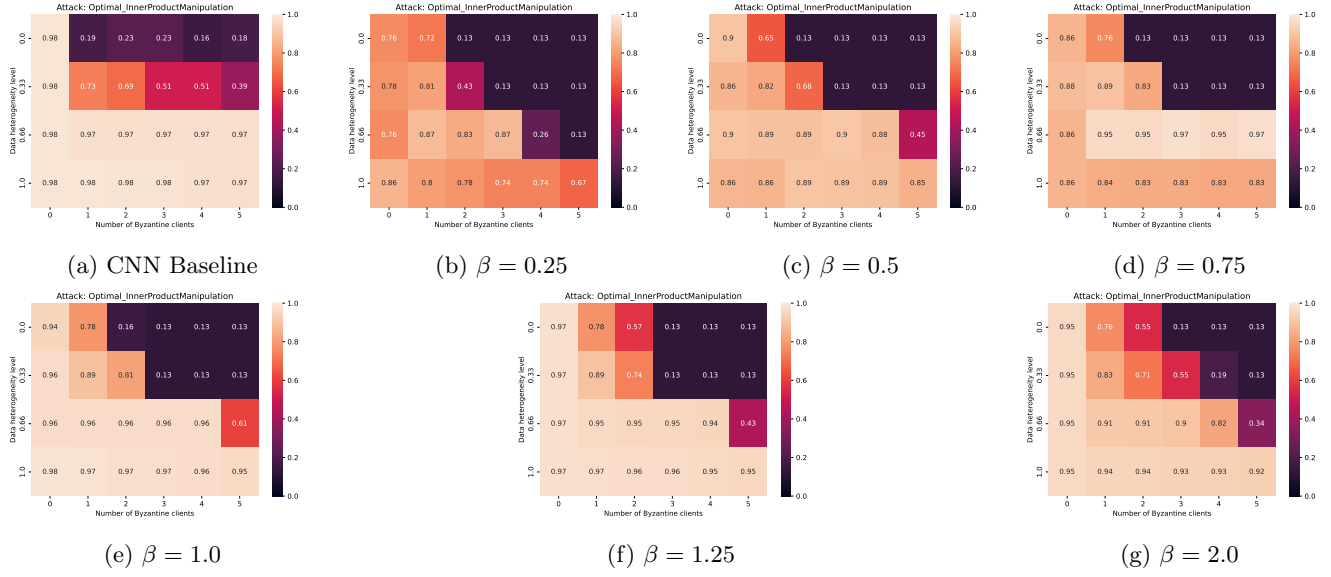


Figure 20: Truncated Mean performance under Optimal Inner Product Manipulation across β values.